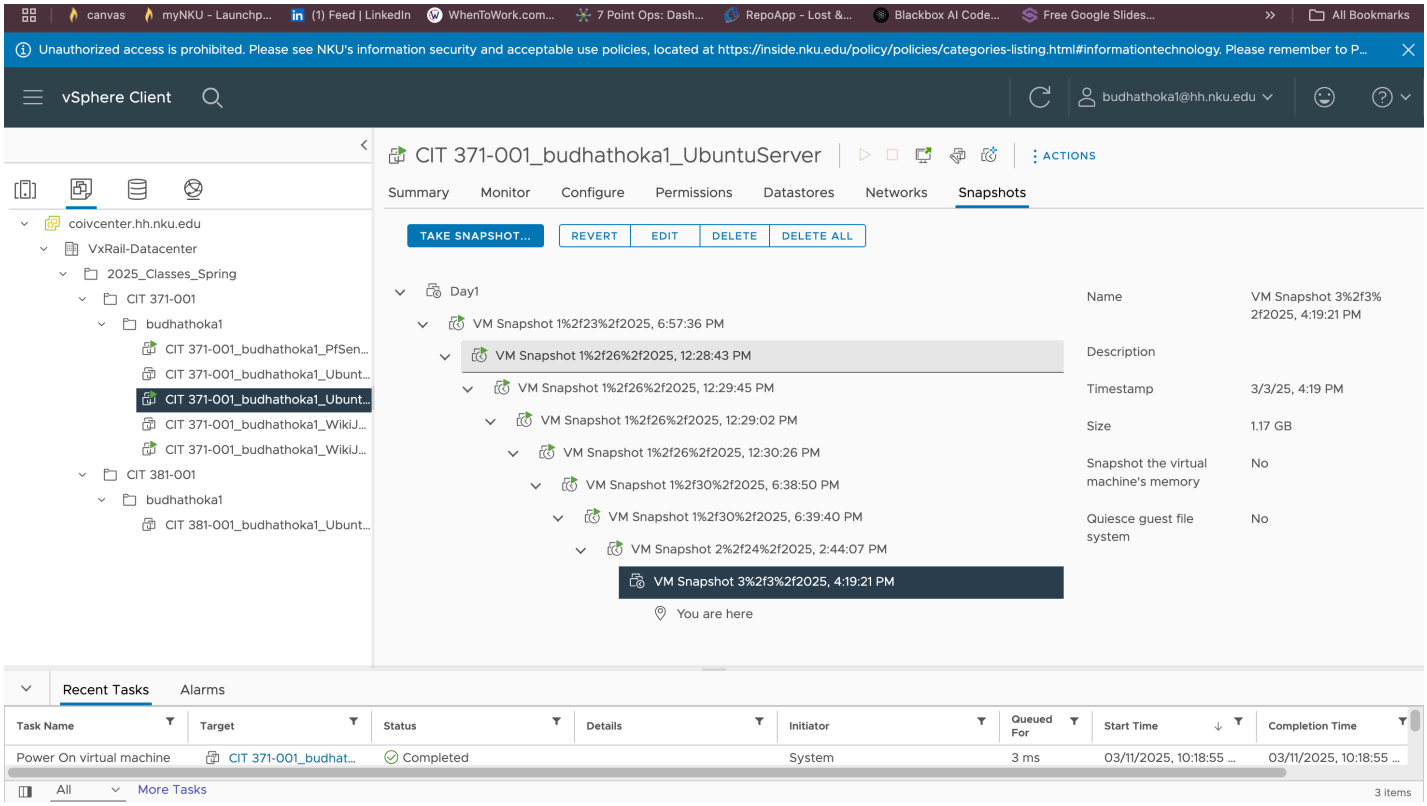
Lab08

Lab 08: Security and Firewall Management

In this lab, we explored techniques to improve the security of an Ubuntu server by performing system security checks, managing firewall configurations, and securing services. Additionally, we worked with GPG encryption and restricted user access.

Part 0: Mandatory VM Snapshots

Before proceeding with the lab, I took a snapshot of the Ubuntu Server VM to ensure that I could revert to a known good state if something breaks. This helps prevent the need for VM replacement requests.



Part 1: Preparing Your Lab Environment

I signed into COIVcenter, turned on the PfSense VM, and then the Ubuntu server. After booting, I logged into the student user account with the provided password. To verify network connectivity, I pinged 8.8.8.8, confirming the server's network setup was functional.

Part 2: Security Checks and Configurations

→ sudo apt update

→ sudo apt upgrade -y

→ sudo useradd testuser

→ sudo passwd -d -u testuser

→ sudo cat /etc/shadow | awk -F: '($2=="!" || $2=="*" || $2==="") {print $1}'

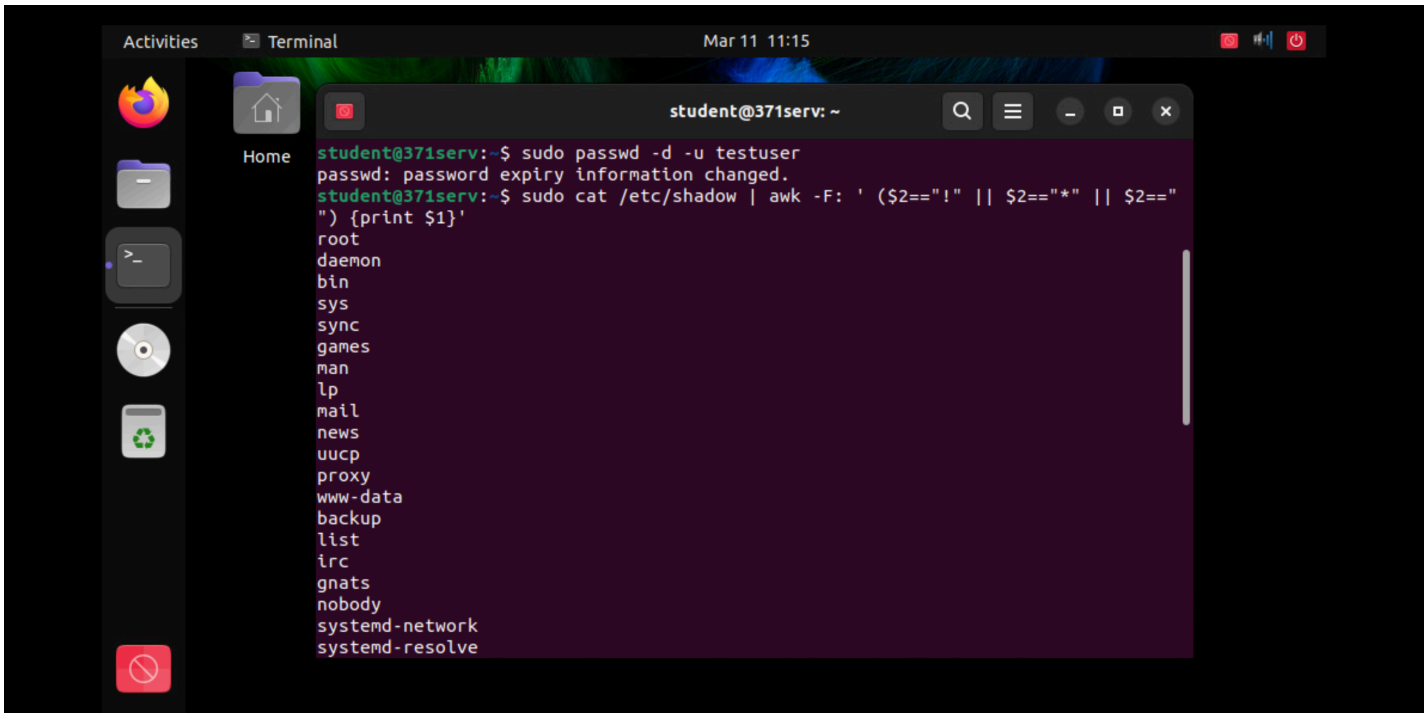
→ sudo userdel testuser

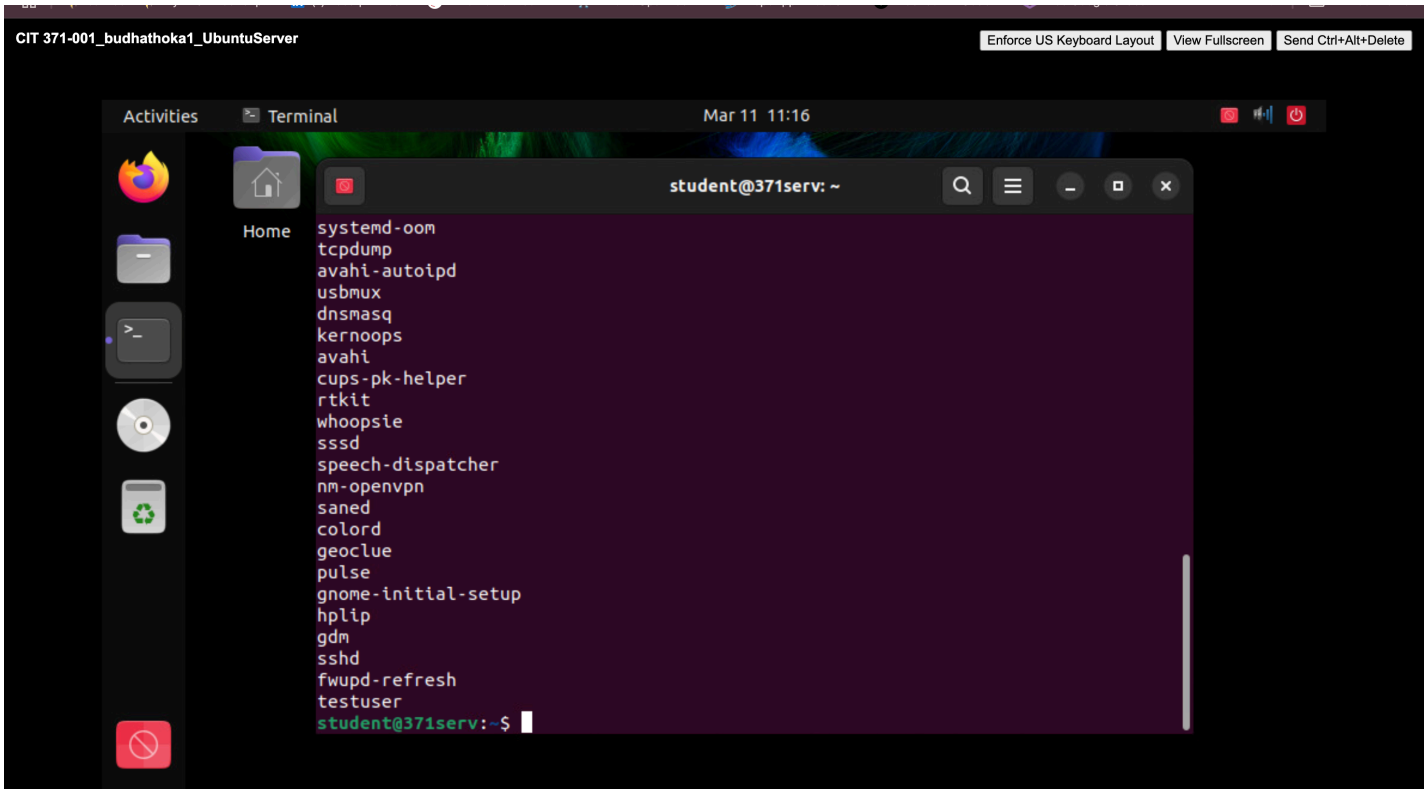
→ echo "Secure Server" > /root/secure.txt

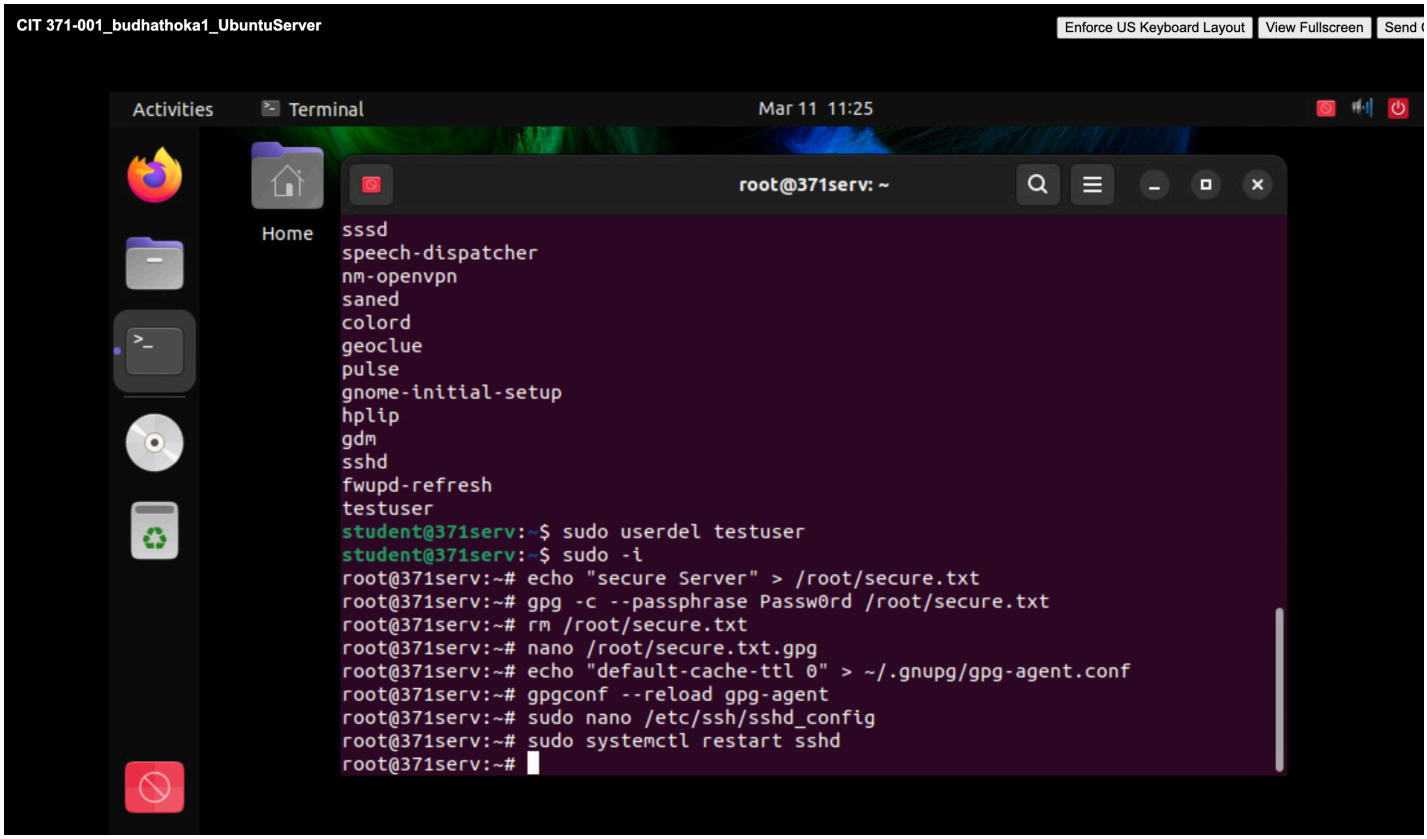
→ gpg -c --passphrase Passw0rd /root/secure.txt

→ rm /root/secure.txt

→ sudo systemctl restart sshd







Part 3: Firewall Checks and Configurations

→ sudo ufw status verbose

→ sudo ufw logging on

→ sudo ufw deny 23

→ sudo ufw insert 1 allow 8080

→ sudo ufw allow from 192.168.1.255 to any port 20 proto tcp

→ sudo ufw allow from 192.168.1.255 to any port 21 proto tcp

```
root@371serv:~# sudo systemctl restart sshd
root@371serv:~# sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)

root@371serv:~#
```

```
root@371serv:~# sudo ufw allow from 192.168.1.255 to any port 20 proto tcp
Rule added
root@371serv:~# sudo ufw allow from 192.168.1.255 to any port 21 proto tcp
Rule added
root@371serv:~#
```

Conclusion:

In this lab, I applied security configurations to an Ubuntu server, improving its defense against unauthorized access and potential vulnerabilities. The firewall configurations restricted access to specific ports, while GPG encryption ensured sensitive data was protected. These tasks reinforce the importance of maintaining server security and managing access control effectively.

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